

Dami Thomas

Cambridge, MA | 678-499-7434 | dvthomas@mit.edu | dvthomas01.github.io/Personal-Website | github.com/dvthomas01

EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA · June 2026

S.B. in Artificial Intelligence and Decision Making, Minor in Mechanical Engineering

- **Relevant coursework:** Introduction to Robotics, Robotic Manipulation, Introduction to Machine Learning, Advances in Computer Vision, Introduction to Autonomous Machines, Dynamics and Controls I & II, Design and Manufacturing

PUBLICATIONS

A. Thomas, R. Galliath, A. Garbuz, L. Anger, C. O'Neill, T. Johst, **D. Thomas**, G. Lordos, J. P. How. "LunarLoc: Segment-Based Global Localization on the Moon." 2025.

RESEARCH EXPERIENCE

MIT Reliable Autonomous Systems Lab (REALM) — Undergraduate Researcher

Cambridge, MA · Aug 2025 – Present

- Trained model predictive control and PPO navigation policies for Spot robots in NVIDIA Isaac Sim industrial environments, advised by Prof. Chuchu Fan.
- Enforced real-time safety using control barrier functions for collision-free navigation.
- Deployed policies across multiple robots with independent waypoint tracking.

NASA Lunar Autonomy Challenge — Team Member

Cambridge, MA · Oct 2024 – May 2025

- Modeled lunar surface elevation with SciPy interpolation (75–90% confidence) and built a PID controller for rover path following and boulder avoidance; contributed to the LunarLoc localization work above.
- Placed 2nd nationally among 31 U.S. university teams for autonomous rover performance.

SELECTED PROJECTS

Locomotion on Compliant Terrain — Independent Research <https://github.com/dvthomas01/compliant-terrain-locomotion>

May 2026 – July 2026

- Trained PPO quadruped locomotion policies on a simulated Unitree Go1 in MuJoCo using a PMTG residual control architecture, running an 8 seed controlled ablation across terrain compliance and observation history to study robustness and performance tradeoffs in legged locomotion.
- Established a clean monotonic robustness ordering across 10 held out terrains over 8 seeds, and refuted three intuitive hypotheses via direct controlled tests, that foot history aids survival, that curriculum progress predicts robustness, and that gait speed causes brittleness, identifying gait contact steadiness as the operative robustness correlate.

Semantic Navigation Robot <https://github.com/dvthomas01/semantic-robot>

Jul – Aug 2025

- Built a semantic visual navigation system with ROS 2 Humble, SLAM Toolbox, Nav2, YOLOv11, and Whisper, integrating perception, planning, and natural-language commands over the NVIDIA R2B dataset.

Voice-Controlled Self-Balancing Rover https://github.com/dvthomas01/voice_rover

Jan 2026

- Designed a self-balancing rover (Fusion 360, Raspberry Pi, ESP32) with a 100 Hz PID balance controller and a real-time voice-to-motor pipeline using wake-word detection and Vosk speech-to-text.

INDUSTRY EXPERIENCE

National Instruments (Emerson) — Embedded Engineering Intern

Austin, TX · June 2026 – Present

- Benchmarked gRPC communication performance between NI PXIe-8881 and NVIDIA DGX Spark hardware, evaluating grpc-direct across hardware and software configurations.

Rockwell Automation — AI Engineer Intern

Milwaukee, WI · Jun – Aug 2025

- Built a RAG re-ranking pipeline for a pharmaceutical recipe-generation product (Streamlit, ChromaDB, LangChain, Azure OpenAI), reaching an 84% user preference rate.

Nasdaq — AI Engineer Intern

Boston, MA · Jan – Feb 2025

- Built a year-end review synthesizer (AWS Bedrock, LLMs, LangGraph) that reduced evaluation-data collection from days to minutes via parallel processing of Jira, email, and GitHub activity.

Elinta Robotics — Assembly and Design Engineer Intern

Kaunas, Lithuania · Jun – Aug 2023

- Designed automated assembly systems with FESTO pneumatics and built a PCB tester integrating pneumatic actuators with computer-vision quality control.

TECHNICAL SKILLS

Robotics & Control: MuJoCo, NVIDIA Isaac Sim, ROS 2, Stable-Baselines3, PPO, MPC, control barrier functions, PID control, Nav2, SLAM Toolbox

Machine Learning: PyTorch, TensorFlow, reinforcement learning, reward shaping, YOLO, CLIP, Hugging Face, NumPy, Pandas, Scikit-learn

Software: Python, MATLAB, Rust, JavaScript, C/Arduino, SQL, Git, Linux, Docker

Hardware & Maker: SolidWorks, Fusion 360, 3D printing, CNC mill and lathe, laser cutter, waterjet, welding